박 사 학 위 논 문
Ph.D. Dissertation

# 이동통신 표준의 보안 취약점을 활용한 이중 용도 기술 연구: 물리적 위치 추적 및 가짜 기지국

# Cellular Dual-Use Technologies Exploiting Specification Vulnerabilities: Physical Localization and Fake Base Stations

2026

오 택 경 (吳 澤 炅 Oh, Taekkyung)

한 국 과 학 기 술 원

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오 택 경

위 논문은 한국과학기술원 박사학위논문으로
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2026년 6월 10일

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# Cellular Dual-Use Technologies Exploiting Specification Vulnerabilities: Physical Localization and Fake Base Stations

Taekkyung Oh

Advisor: Yongdae Kim

A dissertation submitted to the faculty of
Korea Advanced Institute of Science and Technology in
partial fulfillment of the requirements for the degree of
Doctor of Philosophy in Computer Science (Information Security)

Daejeon, Korea
June 10, 2026

Approved by

Yongdae Kim
Professor of Electrical Engineering

The study was conducted in accordance with Code of Research Ethics[1].

[1] Declaration of Ethical Conduct in Research: I, as a graduate student of Korea Advanced Institute of Science and Technology, hereby declare that I have not committed any act that may damage the credibility of my research. This includes, but is not limited to, falsification, thesis written by someone else, distortion of research findings, and plagiarism. I confirm that my thesis contains honest conclusions based on my own careful research under the guidance of my advisor.

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# Curriculum Vitae

## Taekkyung Oh (TK)

Security Researcher
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### BIO

Taekkyung Oh (TK) is a Postdoctoral Researcher in the Department of Electrical Engineering at KAIST. He received his Ph.D. degree in August 2026, advised by Prof. Yongdae Kim. His research interests lie in the area of cellular & satellite network security, wireless system security, and cyber-physical system (CPS) security. Recently, he was a Visiting Scholar at the Florida Institute for Cybersecurity (FICS) Research at the University of Florida. He has published his research in top-tier security and networking venues such as USENIX Security and ACM MobiCom, as well as premier wireless security venues like ACM WiSec. He is also actively involved in practical vulnerability discovery, with his disclosures acknowledged by major vendors and organizations such as Samsung, Qualcomm, MediaTek, and the GSMA.

### EDUCATION

Ph.D. (August, 2026)

- Graduate School of Information Security,
  Korea Advanced Institute of Science and Technology (KAIST), South Korea.
- Supervisor: Prof. Yongdae Kim

M.S. (August, 2021)

- Graduate School of Information Security,
  Korea Advanced Institute of Science and Technology (KAIST), South Korea.
- Supervisor: Prof. Yongdae Kim

B.S. (August, 2019)

- Electronic and Electrical Engineering & Computer Science and Engineering (double major),
  Sungkyunkwan University (SKKU), South Korea.
- Supervisor: Prof. Hyoungshick Kim

### PUBLICATIONS

- Devilray: A Systematic Adversarial Model Revealing Blind Spots in Fake Base Station Detection.
  **Taekkyung Oh***, Duckwoo Kim*, Hansung Bae, Beomseok Oh, CheolJun Park, Tyler Tucker, Nathaniel Bennett, Sangwook Bae, Byeongdo Hong, Patrick Traynor, Yongdae Kim (* Co-first authors).
  *Under review at IEEE ACSAC 2026 and preprinted on arXiv.*

- Denied by Border: Denial-of-Service Attack Exploiting Location Restrictions in Non-Terrestrial Networks.
  Kwangmin Kim, **Taekkyung Oh***, Yongdae Kim* (* Co-corresponding authors).
  *ACM WiSec 2026.*
- LLFuzz: An Over-the-Air Dynamic Testing Framework for Cellular Baseband Lower Layers.
  Tuan Dinh Hoang, **Taekkyung Oh**, CheolJun Park*, Insu Yun*, Yongdae Kim (* Co-corresponding authors).
  *USENIX Security 2025.*
- Passive Three-Dimensional User Equipment Tracking Using Long-Term Evolution Uplink Signals.
  Yooho Jang, Juhong Park, **Taekkyung Oh**, Yongdae Kim, Soeng Ook Park.
  *IEEE Transactions on Instrumentation and Measurement (TIM) 2025.*
- Enabling Physical Localization of Uncooperative Cellular Devices.
  **Taekkyung Oh**, Sangwook Bae, Junho Ahn, Yonghwa Lee, Tuan Dinh Hoang, Min Suk Kang, Nils Ole Tippenhauer, Yongdae Kim.
  *ACM MobiCom 2024.*
- LTESniffer: An Open-source LTE Downlink/Uplink Eavesdropper.
  Tuan Dinh Hoang, CheolJun Park, Mincheol Son, **Taekkyung Oh**, Sangwook Bae, Junho Ahn, Beomseok Oh, Yongdae Kim.
  *ACM WiSec 2023.*
- Void: A Fast and Light Voice Liveness Detection System.
  Muhammad Ejaz Ahmed, Il-Youp Kwak, Jun Ho Huh, Iljoo Kim, **Taekkyung Oh**, Hyoungshick Kim.
  *USENIX Security 2020.*
- Poster: I Can't Hear This Because I Am Human: A Novel Design of Audio CAPTCHA System.
  Jusop Choi, **Taekkyung Oh**, William Aiken, Simon S. Woo, Hyoungshick Kim.
  *ACM ASIACCS 2018.*
- Hey Siri – are you there?: Jamming of voice commands using the resonance effect.
  **Taekkyung Oh**, William Aiken, Hyoungshick Kim.
  *IEEE ICSSA 2018.*

## RESEARCH PROJECTS

- Development of a Fake Base Station Detection System in Cellular Networks (October, 2025 – April, 2026).
  Funded by the Global Advanced Cybersecurity Human Resources Development program, South Korea ($24K).
  Visiting research at the University of Florida
  *Principal Investigator.*
- Developing a Technology for Bypassing Fake Base Station Detection Methods (March, 2025 – December, 2025).
  Funded by the Korean National Intelligence Service (NIS) ($100K).
  *Project manager.*
- Developing a Technology for Detecting an Open-Source Based Fake Base Station (February, 2024 – December, 2024).
  Funded by the Korean National Intelligence Service (NIS) ($100K).
  *Project manager.*
- Tracking and Identifying Devices and Call Traffic in Voice Phishing Ecosystem (April, 2022 – December, 2024).
  Funded by the Korean National Police Agency ($4.5M).
  *Project manager.*

## INVITED TALKS

- Testing, Analyzing, Improving Security of Cellular Networks with srsRAN (October, 2023).
  srsRAN Project Workshop, Software Radio Systems (SRS). Arlington, Virginia, USA.
- Voice Phisher Intelligence: Location Tracking and Infrastructure Vulnerability Analysis (November, 2022).
  Security Intelligence Summit (SIS), S2W Inc. Seoul, South Korea.

## Security Reports & Acknowledgements

- GSMA CVD-2026-0115: Specification vulnerability in the location restriction mechanism in Non-terrestrial networks (NTN).
- CVE-2025-20708: Memory corruption in MediaTek modem chipsets.
- CVE-2024-48883: Information leakage in Samsung and Google modem chipsets.
- CVE-2024-23385: Memory corruption in Qualcomm modem chipsets.
- CVE-2024-20077: Memory corruption in MediaTek modem chipsets.
- CVE-2024-20076: Memory corruption in MediaTek modem chipsets.
- GSMA CVD-2023-0077: Specification vulnerability in the uplink scheduling mechanism in LTE networks.
- GSMA CVD-2023-0070: Specification vulnerability in the radio control mechanism in LTE networks.

## Honors & Awards

- 2025
  Outstanding Student Researcher Award, Telecommunications Technology Association (TTA) — Chair's award.
- 2024
  Student Travel Grant for ACM MobiCom 2024, ACM SIGMOBILE.
- 2019 – 2026
  Korean Government Scholarship – Full-ride scholarship for M.S. and Ph.D. programs.
- 2018
  Undergraduate Research Conference, College of Software, Sungkyunkwan University – Bronze prize.
  Dean's List, Sungkyunkwan University – Reward for students with outstanding academic performance (Top 5%).
- 2017
  Global Idea Contest, Daewoo Electronics – Development of noise canceling technique for home appliances, Honorable mention award.
  Dean's List, Sungkyunkwan University – Reward for students with outstanding academic performance (Top 5%).
- 2016
  Idea Contest of Assistive Technology Device for Disabled Students, Korean Ministry of Education – Development of function-graph converter for the blind, First prize (Minister's award).
  Jinggeomdari Scholarship, Sungkyunkwan University – Scholarship for students with outstanding academic performance.

## Experience

- Visiting Scholar, University of Florida (October, 2025 – April, 2026).
  Conducting research on cellular network security at the Florida Institute for Cybersecurity (FICS) Research, advised by Prof. Patrick Traynor.
  Supported by the Global Advanced Cybersecurity Human Resources Development program, South Korea.
- Counseling Assistant, KAIST (February, 2024 – February, 2025).

Conducting over 30 student counseling sessions for graduate life and research.

- External Security Advisor, Republic of Korea Marine Corps (April, 2023 – December, 2024).
  Analyzing global trends in cyber warfare techniques and strategies.
- External Reviewer (2020 – 2025).
  IEEE Symposium on Security and Privacy (SP 2024).
  USENIX Security Symposium (Security 2023).
  Network and Distributed System Security Symposium (NDSS 2020–2021, 2025).
- Teaching Assistant, KAIST (September, 2020 – August, 2022).
  (EE488) Introduction to Cryptography Engineering | 2022 Spring.
  (EE515) Theory of Hacking (Security 101) | 2021 Fall.
  (CS448) Introduction to Information Security | 2021 Spring.
  (IS521) Information Security Laboratory | 2020 Fall.
- External Software Advisor, Republic of Korea Ministry of Education (December, 2019).
  Consulting on the development of braille devices and assistive applications for visually impaired individuals.
- Undergraduate Researcher, SKKU Security Lab (December, 2016 – December, 2018).
  Conducting research on usable security and voice authentication, advised by Prof. Hyoungshick Kim.
- Web Developer, SKKU System Consultant Group (SCG) (March, 2016 – July, 2017).
  Full-stack developer for web applications (JavaScript, SQL, HTML, CSS).
- Combat Engineer, Republic of Korea Army (October, 2013 – June, 2015).

## Languages & Skills

- Korean (Native)
- English (Upper intermediate)
- Python and CUDA
- C, C++
- LaTeX
- Web development (JavaScript, SQL, HTML, CSS)